FMI	ADMINISTRATIVE INFORMATION	FMI No: 25382

FMI Title: **FMI 25382 - VCT Rotating 48V PS Fuse Board replacement and Torque label update.**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☑ Mandatory Safety	GEMS - AM:	Issue Date: 9/1/2006
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date: 3/16/2007

☑ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

☑ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

☐ **Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

☐ **No Parts** are required to complete this FMI.

☐ **Batch with** _____

To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).

☐ **Non-Staged** **No Customer List is provided**. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

Unit Tracked: 5124069

FMI Instructions Part No.: 5174417-100

Do NOT leave unsecured on site

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FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25382 **FMI Title:** FMI 25382 - VCT Rotating 48V PS Fuse Board replacement and Torque label update.

Customer Full Name: _____

Number and Street Address: _____

City and Country: _____

_____

Country Code **Site Number** **System I.D.**

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

_____

_____

Installer Comments _____

Effectivity

The following CT Equipment is affected:

All VCT systems manufactured in the Milwaukee MFG before January 2006

All VCT and LS Pro 32 systems manufactured in the Hino MFG before July 2006

Model number tracked: 5124069 VCT Gantry & 5123200 Pro 32 Detector

Purpose

The purpose of this FMI is to update the VCT installed base with the redesigned rotating 48V fuse board. The change provides more reliability to the connectors of the board and adds a ferrite to the power supply cables to minimize low signal noise. This FMI also delivers new torque labels for the rotating assembly. Some sites will have them and others will not, please install the labels as instructed.

Prerequisite FMIs

FMI Number	Title
None	

Related FMIs

FMI Number	Title
None	

Furnished Materials

FMI 25373 Collector 5175587

Qty	Description	Part #
1	FMI 25382 Document	5174417-100
1	Fuse board assembly with ferrite	5167803
1	Torque label for 48V Power Supply to Das Bracket	5155151-3
1	Torque label for Rotating Base to Detector Bottom Rail	5155151-6
1	Torque label for Collimator Base to Rotating Base	5155151-7
6	Torque label for Slip Ring Mounting Casting to Slip Ring	5155151-14
6	Torque label for Slip Ring Mounting Casting to Rotating Base	5155151-15
1	Torque label for 180 Balance Weights	5155151-17
1	Torque label for 107 Balance Weights	5155151-18

Qty	Description	Part #
2	M4 bolt	46-328417P3
2	P-clamp	46-208718P7
1	M4X0.7 Tap	5135684

Tools Required

Qty	Description	SW Version	Part #
1	FMI Document		5174417-100

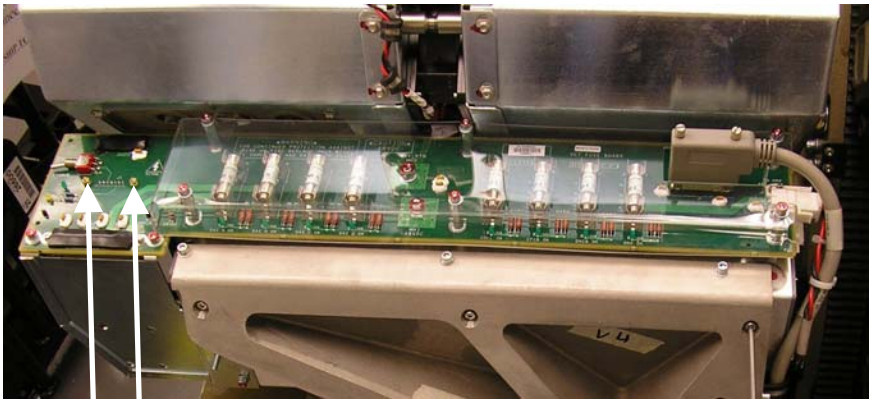
Personnel Requirements

"2" hours labor and "1" hour of travel

1. Initial system inspection to verify if the system needs this FMI

Verify your system's fuse board

New Fuse board is at Assembly Version E or later.



The New Fuse board has connectors that are bolted through the circuit board. The old board had connectors without mechanical hardware support through the board. If your system has the Version E or later board, connectors with mechanical hardware for support and the ferrite on the black and Red power cables, the system does not need the fuse board update.

If your system has the torque labels on the rotating assembly that match the labels from this kit and your fuse board is updated, your system does not need FMI 25382. Close the FMI as "Code 2". Please return the FMI kit to HM Product solutions.

2. VCT Rotating 48V PS Fuse Board Update

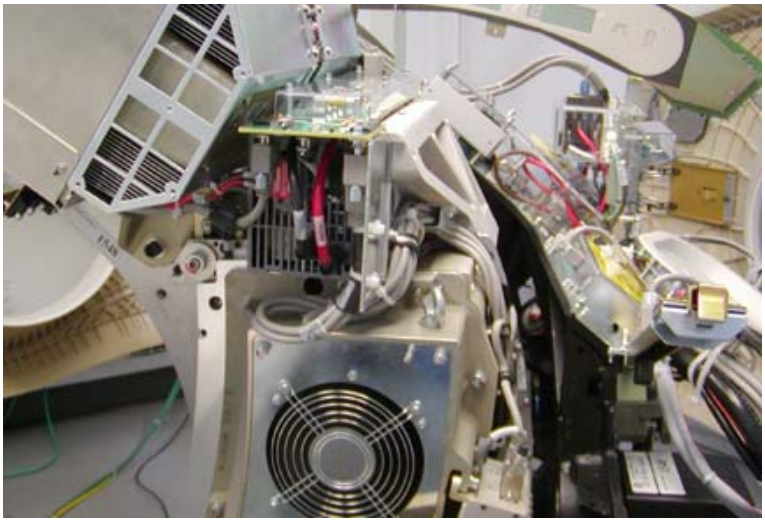
Power Supply Assembly Removal

1. Remove gantry side covers and disable Axial drive from the Service Switch Panel.

   Refer to *Parts Replacement → Gantry → Enclosure → (Cover Removal Procedure).*

2. Rotate the gantry to position the 48V power supply on the right side of the gantry above the service switch panel and lock gantry in position with rotational lock.

Illustration 1: 48V Power Supply Assembly



DANGER

ELECTROCUTION HAZARD

HIGH VOLTAGES PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS. SEE SAFETY MENU OF SERVICE DOCUMENT GUI FOR APPROPRIATE PROCEDURES.

3. Turn off the gantry HVAC and 120 VAC switch on the service panel. Power down the console and use LOTO procedures to make sure gantry power is removed.

4. Remove the gantry front cover.

5. Remove the Power supply output cables on the front and rear of the supply (J1 through J5) as shown in Illustration 2 and Illustration 3. Tyraps for the front cable support need to be removed as shown. Make note of any other tyraps removed during supply removal for reinstallation later.

Illustration 2: 48V Power Supply Front View

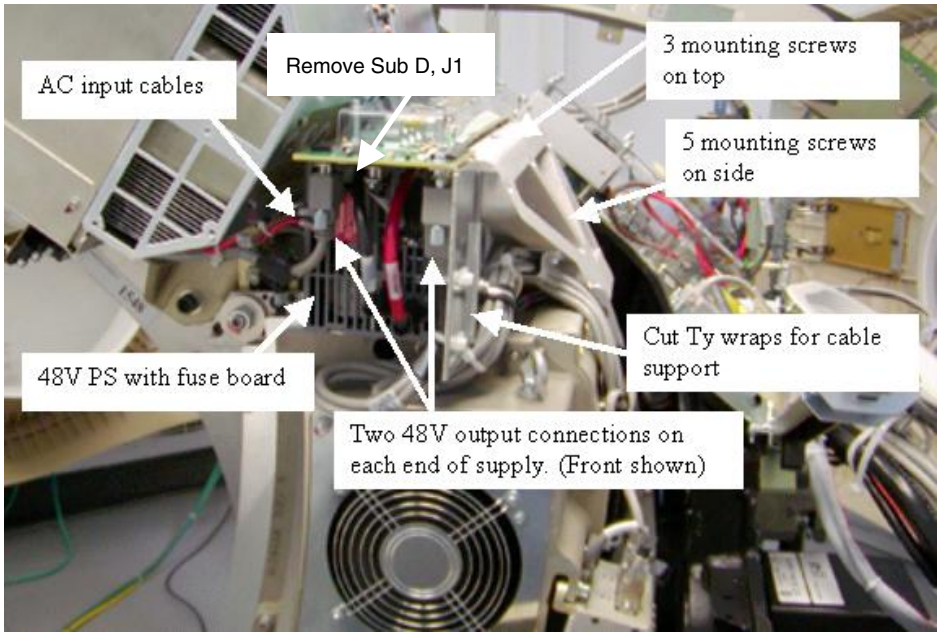
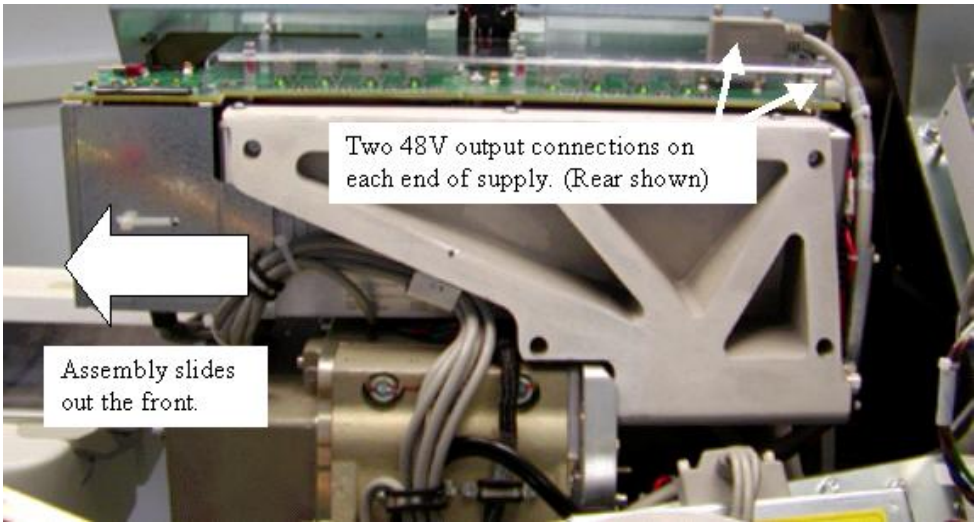


Illustration 3: 48V Power Supply Side View



6. Disconnect the AC input cables from the terminal strip on the front of the power supply.

7. Remove the 8 screws attaching the power supply assembly to the cast frame and slide the supply out the front of the gantry. (The power supply is removed from the Gantry to properly perform the visual inspection in step 3 of “Installing New Fuse Board” section.)

8. Cut the heat shrink on the 48V output cables (Large diameter Red and Black cables.). Remove the hardware that mounts the cables to the 48V supply. (See Ill. 4)

Illustration 4: 48V Power Cables

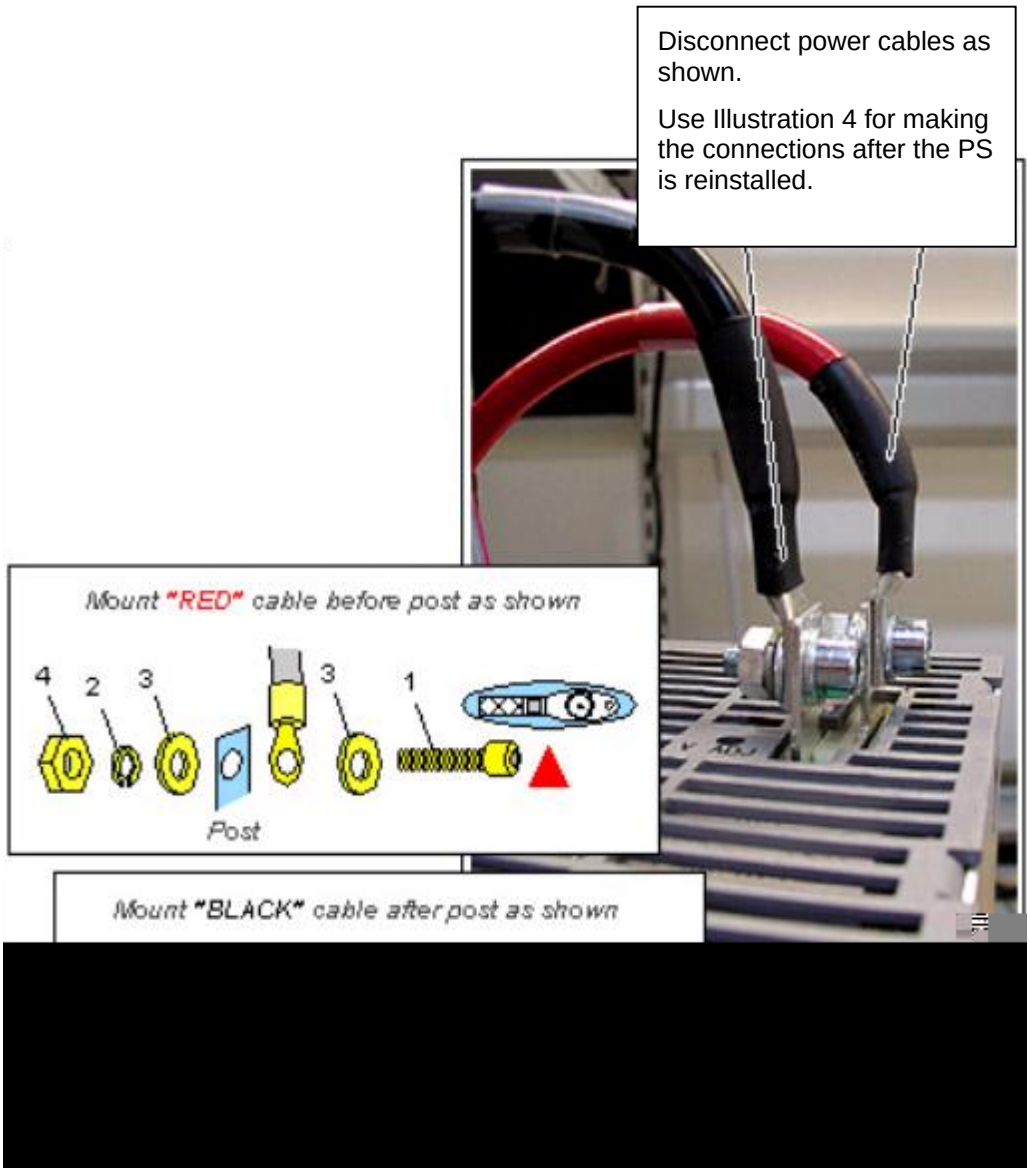
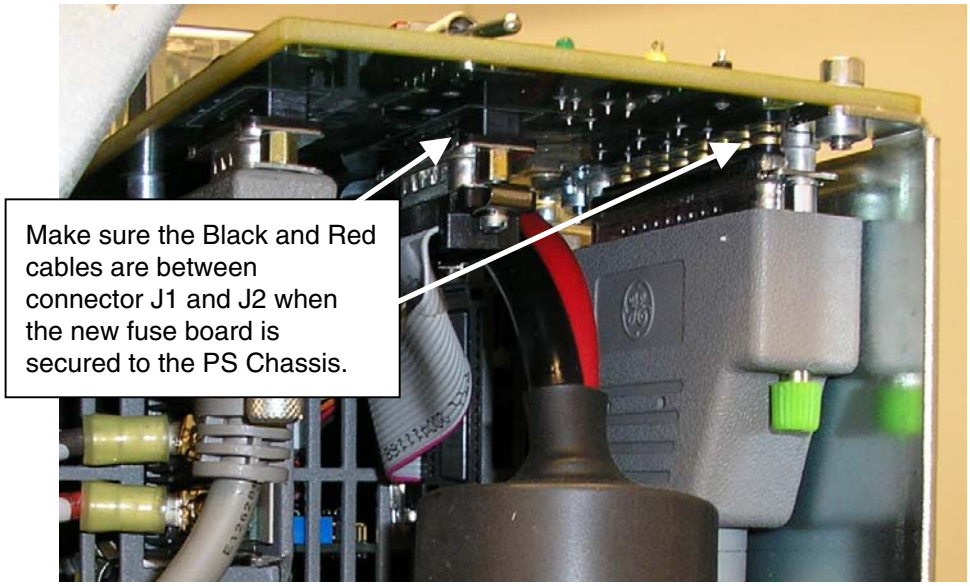
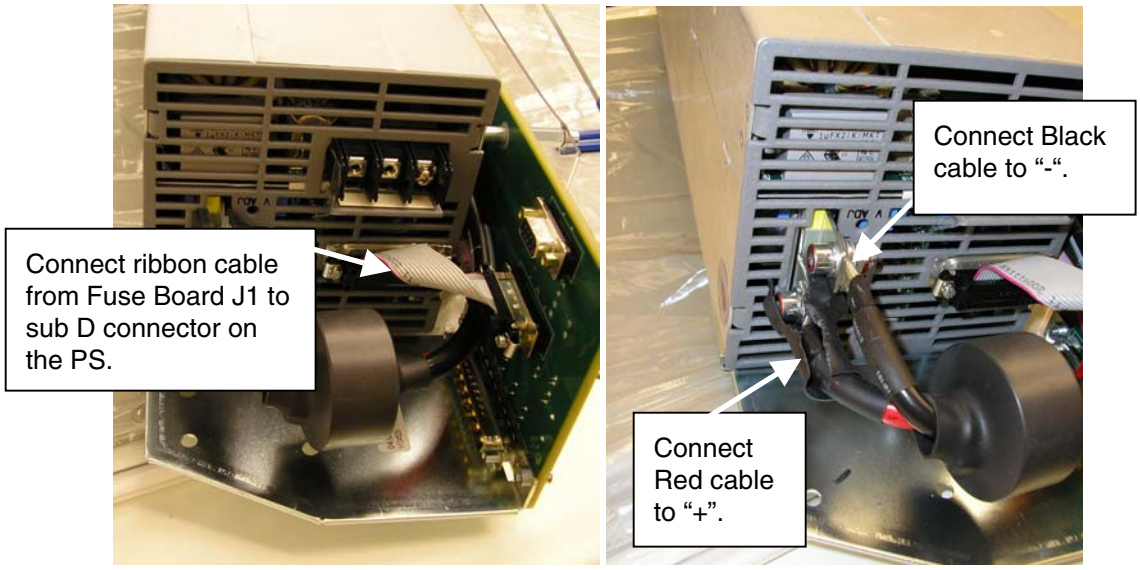


Illustration 7



4. Install the new board (5113714 Rev E) onto the Power supply chassis. Install the six standoffs, torque to 2.3 N-m (1.7 LB-ft, 20.4 LB-in). Make sure the Black and Red cables are between connector J1 and J2 when the new fuse board is secured to the PS Chassis. (See Ill. 7)

Illustration 8



5. Connect the ribbon cable from J1 to the PS sub D connector. (See Ill. 8)

6. Connect the Black cable to "-" and the Red cable to "+" on the power supply using the original hardware. Torque to 7.9 N-m (5.8 LB-ft, 70 LB-in) (See Ill. 4 and 8) - **IMPORTANT: Ensure that the Black cable is connected to the "-" terminal and the Red cable is connected to the "+" terminal. If this is not done properly, damage will result to the DAS/Detector upon powering.**

Illustration 9

7. Check continuity between Fuse board and Power Supply. **IMPORTANT. DAMAGE CAN BE DONE TO DAS DETECTOR IF CABLES ARE NOT HOOKED UP PROPERLY**.
    - Using a DVM, place probe on fuse board end (48V_RTN - MH2) and place probe on (-) terminal on Power Supply End. Ensure continuity between these two points
    - Take DVM and place probe on fuse board end (MH1 - 48VDC) and place probe on (+) terminal on Power Supply End. Ensure continuity between these two points.
    - If continuity checks fail, check power cable connection
      48V_RTN – MH2 (Black cable) connects to " – " terminal on PS.
      48VDC – MH1 (Red cable) connects to " + " terminal on PS.
8. After connecting the Red and Black PS cables, slide the rubber grommet over the bolts holding the cables to the PS. Secure with a tyrap (See Ill. 9).
9. Install the fuse board cover and the six 4mm socket head cap screws, lock washers, and flat washers. Torque to 2.3 N-m (1.7 LB-ft, 20.4 LB-in).

Power Supply Assembly Installation

1. Slide the new power supply assembly in from the front of the gantry.
2. Install the 8 - M4 screws to attach the supply to the cast frame.
3. Torque the 8 - M4 screws to the following value:

2.3 N-m	1.7 lb-ft	20.4 lb-in	23.5 Kg-cm

4. Reconnect input and output cabling to power supply assembly as removed. Install Ty-raps in all locations removed during supply removal.

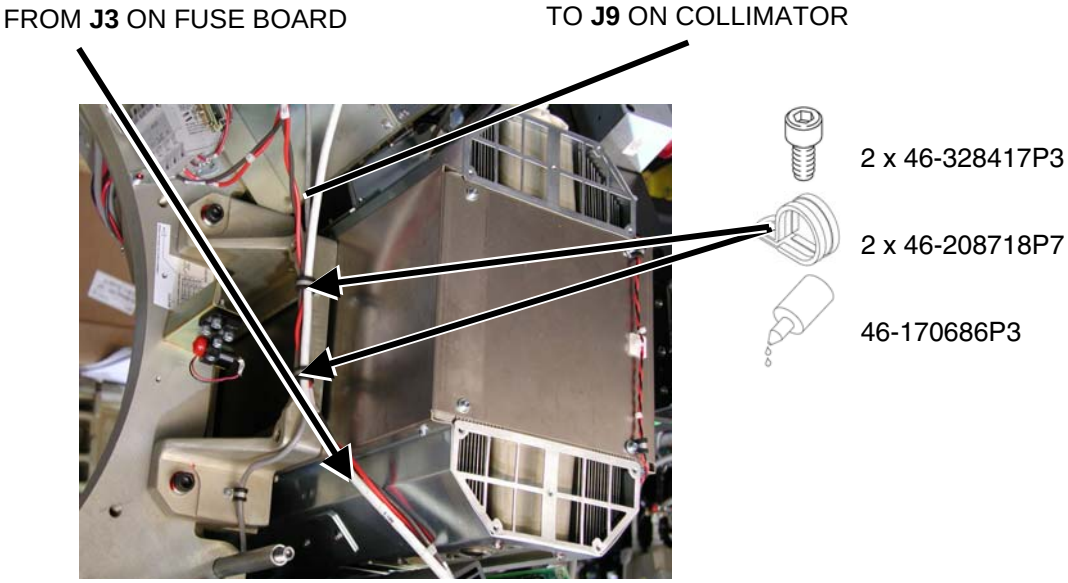
3. VCT P-Clamp Screws

Problem

Screws attaching the collimator cable p-clamps to the HV tank interface bracket may be missing Loctite, which could cause the screws to loosen or fall out. Also, the holes for the screws may not be tapped deep enough.

Solution

- Inspect the screws attaching the p-clamps to the HV tank interface bracket. If there are red marks on the heads, the screws have Loctite and no further action is required for the screws.
- If the screws do not have red marks, or if there are washers installed, remove the screws and washers.
- Confirm the hole is tapped deep enough for the screw to tighten against the casting without washers installed. If not, tap to at least 14mm with an M4 x 0.7 tap.
- Apply Loctite 242 (blue) and reinstall the screws without the washers. Torque to 2.3 N-m (1.7 LB-ft, 20.4 LB-in). See figure.



4. Gantry Torque Label Updates

Some systems may not have torque labels to replace. In these cases, please install the new labels provided in the FMI Kit as indicated in the instruction.

1. Replace label 5131694-3 with 5155151-3 by placing new label directly over existing label on DAS end bracket (fig. 1 & 2)

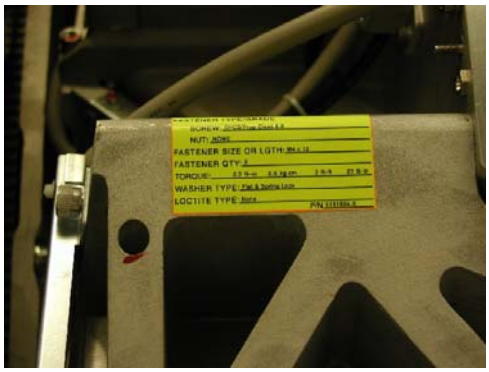


Figure 1: Close-up of label 5131694-3 (48V PWR SUPPLY to DAS BRKT)

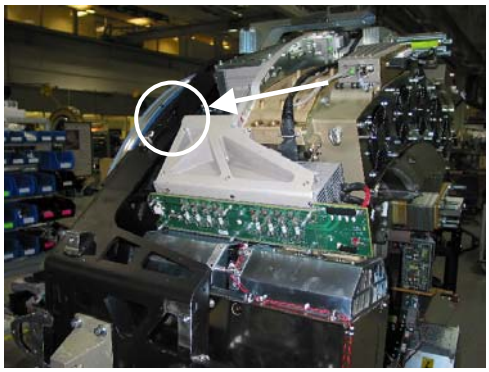


Figure 2: Label 5131694-3 Location on VCT gantry

2. Replace label 5131694-6 with 5155151-6 by placing new label directly over existing label on rotating base near detector (fig. 3 & 4).

Figure 3: Close-up of label 5131694-6 (ROT BASE into DET BOTTOM RAIL)

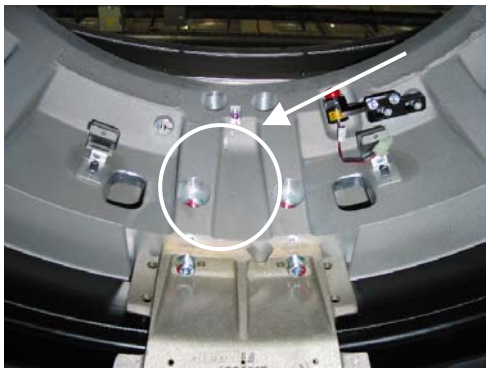


Figure 4: Label 5131694-6 location on VCT gantry

3. Replace label 5131694-7 with 5155151-7 by placing new label directly over existing label on rotating base near tube collimator (fig. 5 & 6).

Figure 5: Close-up of label 5131694-7 (COLL BASE to ROT BASE)

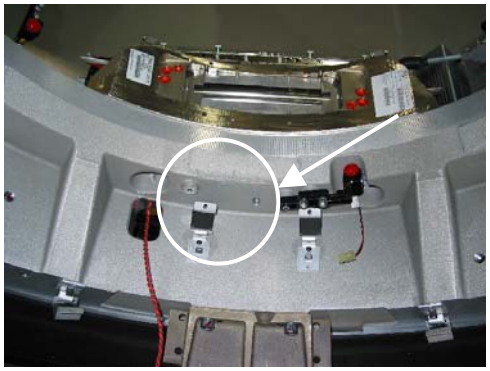


Figure 6: Label 5131694-7 location on VCT gantry

4. Replace labels 5131694-14 and 5131694-15 with 5155151-14 and 5155151-15, respectively, by placing new labels directly over existing labels on slipring mounts (fig. 7,8 & 9) (Qty. 6).

Figure 7: Close-up of label 5131694-15 (S/R MNT CASTING to ROT BASE), *5131694-14 not shown*

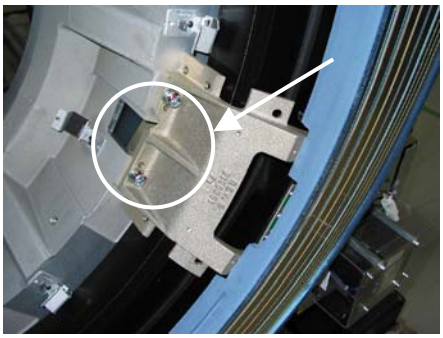


Figure 8: Label 5131694-15 location on VCT gantry

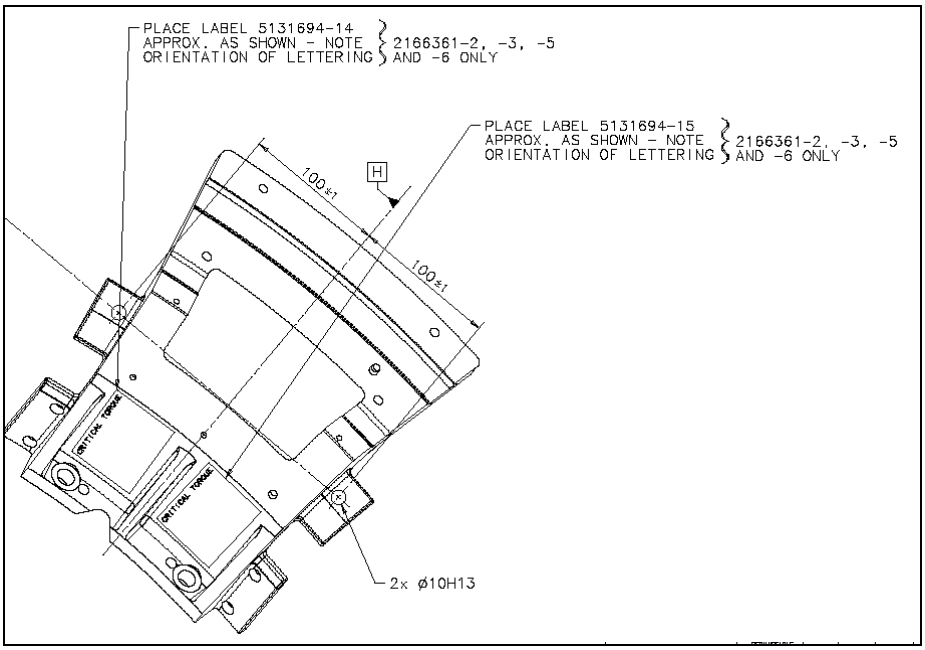


Figure 9: Drawing showing label 5131694-14 and 5131694-15 locations

5. Replace labels 5131694-17 and 5131694-18 with 5155151-17 and 5155151-18, respectively, by placing new labels directly over existing labels on 180 and 107 failsafe rods. There are systems in the field without the failsafe rod in the center of the threaded rods. If you have one of these systems, the label replacement is not applicable (fig. 10 & 11).

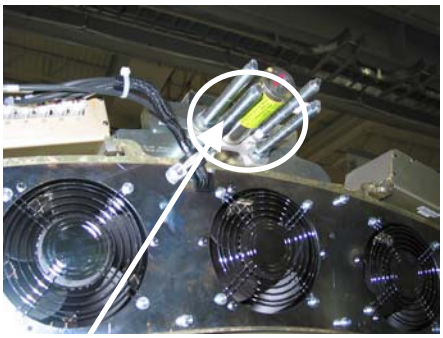


Figure 10: Label 5131694-17 (180 BALANCE WEIGHTS) location on gantry

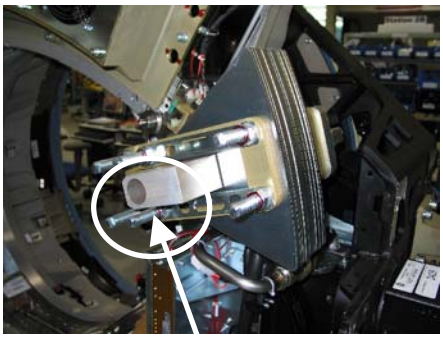


Figure 11: Label 5131694-18 (107 BALANCE WEIGHTS) location on gantry. (Apply label to side of fail safe rod)

5. Finalization

Check power supply output levels and final test scan per *48V Power Supply Adjustment*.

Path to docs:

- LightSpeed 7.X service documentation
- Setup, Cals
- Gantry
- DAS and Detector
- 48V Power Supply Adjustment

6. FMI Closure

Close FMI per regional procedure.

Remove this FMI document from the customer site.

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